Assignment 2: Fuzzification and Defuzzification: 27:08:2026

(Date of Submission: 03-09-2026)

General Instructions

- Write a separate Python program for each question.
- Matrix dimensions, membership values, learning rates, number of epochs, etc. should be accepted as input wherever applicable rather than being completely hard-coded.
- For fuzzy sets and fuzzy relations, membership values must lie in the interval [0, 1].
- For crisp relations, membership values must be restricted to 0 or 1.
- Display the input and the corresponding output clearly.
- Do not use ready-made Fuzzy Logic or Neural Network libraries for implementing the core algorithms.

1. Implement a program to convert a given crisp input into appropriate membership values using user-defined fuzzy membership functions.
2. Implement all the defuzzification methods discussed in class:
   - o Max-Membership / Height Method
   - o Centroid Method
   - o Weighted Average Method
   - o Mean-Max Membership / Middle-of-Maxima Method
   - o Centre of Sums Method
   - o Centre of Highest/Largest Area Method
   - o First of Maxima o Last of Maxima
3. Apply all applicable defuzzification methods to the same fuzzy output set and display the crisp values obtained from each method.
4. Display a comparison of the results and identify whether different methods produce the same or different crisp outputs.